

CHAPPIDI RAMA PRAKASH

Full Stack Developer | Backend Engineer | Cloud & Real-Time Systems

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PROFESSIONAL SUMMARY

Results-driven Full Stack / Backend Developer with hands-on experience designing and deploying scalable, production-grade systems. Proficient in RESTful APIs, real-time platforms, and cloud-hosted services using Node.js, Express.js, PostgreSQL, Redis, Socket.IO, and AWS. Passionate about clean architecture, performance optimization, and delivering reliable systems at scale. Solved 200+ LeetCode problems with strong focus on Graphs and Linked Lists.

TECHNICAL SKILLS

Languages	JavaScript, Java, C
Backend	Node.js, Express.js, REST APIs, MVC Architecture, RBAC, JWT, Middleware Design
Real-Time	Socket.IO, Live GPS Tracking, Bidirectional Communication, Event-Driven Architecture
Databases	PostgreSQL, Redis — Schema Design, Query Optimization, Indexing, Caching, Stored Procedures
Cloud & DevOps	AWS EC2, AWS S3 (presigned URLs), Nginx (reverse proxy, SSL), PM2
Frontend	React, Vite, HTML5, CSS3, html2canvas, jsPDF
Data & Tools	Pandas, Matplotlib, BeautifulSoup, Git, GitHub, Postman, Linux CLI

PROFESSIONAL EXPERIENCE

Freelance Full Stack Developer | Vegaa – Real-Time Booking Platform

Feb 2026 – Apr 2026

Node.js, Express.js, Socket.IO, Redis, PostgreSQL, AWS EC2, Nginx

- Designed and built a real-time on-demand service booking platform (Uber/Rapido-inspired) connecting users with nearby providers with live status updates and GPS tracking throughout the session lifecycle.
- Implemented bidirectional event-driven communication using Socket.IO — handling booking requests, provider acceptance/rejection, state transitions (requested → accepted → en-route → completed), and in-app messaging.
- Built live GPS location tracking with continuous coordinate emission from provider to user, enabling real-time map updates and ETA estimation.
- Leveraged Redis for session management, socket state persistence, and pub/sub horizontal scaling — ensuring correct event routing across multiple server instances.
- Engineered provider matching logic using optimized PostgreSQL spatial queries to rank nearby providers by geolocation, service type, and availability.
- Deployed on AWS EC2 behind an Nginx reverse proxy with WebSocket upgrade support for persistent socket connections in production.

Freelance Full Stack Developer | CoreFix MEP Services

Sep 2025 – Jan 2026

Node.js, Express.js, PostgreSQL, AWS EC2, AWS S3, JWT, REST APIs

- Architected and developed a multi-module backend system for a Mechanical, Electrical & Plumbing (MEP) service company — handling service booking, AMC subscription management, user roles, and technician job assignment.
- Designed normalized PostgreSQL schemas supporting complex relationships between customers, service requests, technician profiles, job statuses, and billing records.
- Built secure REST API endpoints with JWT-based authentication and RBAC separating Admin, Technician, and Customer permissions across all routes.
- Deployed on AWS EC2: configured Ubuntu server, Nginx reverse proxy, SSL, and PM2 process management for zero-downtime restarts.
- Integrated AWS S3 for scalable media storage with pre-signed URLs for secure, time-limited upload and retrieval of service reports and documents.

Database Intern | Algonox Pvt. Ltd.

Oct 2024 – Mar 2025

PostgreSQL, MSSQL, Database Management & Backend Integration

- Executed a full-scale data migration from Microsoft SQL Server (MSSQL) to PostgreSQL, covering schema translation, data type mapping, and post-migration validation — ensuring zero data loss and consistent query performance.
- Designed and authored SQL scripts, stored procedures, and triggers to automate repetitive data operations, reducing manual overhead for backend teams.

- Participated in schema normalization and relational database design applying 3NF principles to minimize redundancy and support scalable backend services.
- Collaborated with backend developers to integrate optimized database layers with RESTful API endpoints, improving response times and maintainability.
- Conducted performance profiling of slow-running queries using EXPLAIN ANALYZE and applied indexing strategies to improve query execution plans.

PROJECTS

Invoice Generator — React PDF Builder

2026

React, Vite, CSS, html2canvas, jsPDF, LocalStorage

- Client-side invoice builder with live preview, dynamic line-item calculations (subtotal, GST, discount, grand total), logo/QR/signature uploads, and one-click PDF export via html2canvas + jsPDF.

YouTube Channel Analysis – Web Scraping & Data Viz

2025

Python, BeautifulSoup, Pandas, Matplotlib, Seaborn

- Scraped YouTube metrics (views, likes, engagement), cleaned data with Pandas, and visualized trends via charts and heatmaps to surface actionable content performance insights.

DSA & COMPETITIVE PROGRAMMING

Solved 200+ LeetCode problems (Easy, Medium & Hard) with strong proficiency in Graphs (BFS, DFS, topological sort, cycle detection), Linked Lists, Trees, Sliding Window, Two Pointers, Backtracking, and Dynamic Programming. Consistent focus on time/space complexity analysis.

EDUCATION

B.Tech – Computer Science & Engineering | RGUKT IIIT
Srikakulam

June 2021 – May 2025

CGPA: 8.2 / 10.0 | Coursework: DSA, DBMS, OS, Computer Networks, Software Engineering, OOP

Pre-University Course (PUC) | Board of Intermediate Education,
AP

Aug 2019 – Apr 2021

CGPA: 8.9 / 10.0

CERTIFICATIONS

- SQL Certification — Great Learning | 2024
- MongoDB Certification — MongoDB University (Official) | 2023